# Which tag pair should an indie game use? First-Person × Multiplayer

MADT 7202 Mid-Term · Panjapol Ampornratana 6810424009 · Repo: github.com/hhej/steam-tag-whitespace

Steam is the largest PC game store. Players vote tags onto each game; the top tags drive store search.

## 1 The decision

An indie studio's creative director picks the next game's two lead tags; if no pair passed the selection rule, they would pick a crowded one (table, last row).

## 2 Why a network

Which tags sit in separate neighbourhoods, and which tag links them? That needs neighbours and paths, which a table of counts lacks.

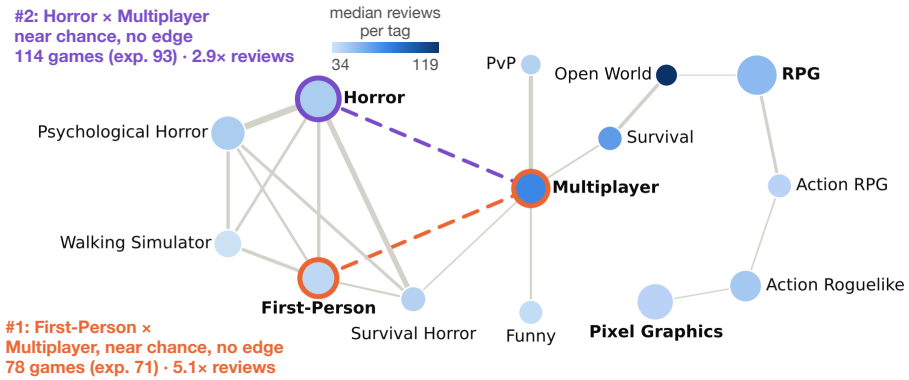
## 3 Data

| | |
|---|---|
| **Source** | S1 FronkonGames steam-games-dataset (Hugging Face); S2 live Steam store. |
| **Obtained** | S1 download (Sep 2025). S2: 1,000 random S1 games (random seed 7202), scraped 2026-10-14. |
| **Legal** | S1 MIT licence. S2: robots.txt allows /app/, /appreviews/; 1 req/1.5 s; no personal data. |
| **Volume** | S1: all 9,094 Indie games released 2023–24 with ≥10 reviews; hold-out: 2,076 released Jan–Jun 2025. S2: 996 with a live page. |
| **Check** | S1 vs S2: mean top-5 tag Jaccard 0.87; Spearman ρ of reviews 0.98. |
| **Bias** | Favours games that sold; reviews = buyers, not unmet demand; late-2024 games had less time. |

## 4 Graph construction

| | |
|---|---|
| **Node** | A Steam tag. Undirected, weighted tag–tag projection of the game–tag bipartite graph. |
| **Edge** | Co-occurrence in a game's top 5 tags; suggests relatedness, not a stated relation. |
| **Resolution** | Co-op, Online/Local Co-Op, Local Multiplayer → Multiplayer (84% of Horror × Multiplayer games were co-op too). |
| **Weight** | NPMI: co-occurrence vs chance (0 = chance). By raw count, 9 of the top-10 edges touch the 5 commonest tags; by NPMI, 0. |
| **Window** | Top 5 tags per game, as the store page shows; with all 20, #1's reviews drop from 5.1× to 3.0×. |
| **Excluded** | "Indie"; Early Access, Free to Play, adult tags; on >25% of games: Singleplayer, Adventure, Action, Casual, 2D, 3D, Atmospheric, Exploration. |
| **Pruning** | Tag on ≥1.5% of games (136); edge needs NPMI ≥ 0.10 and ≥0.5% of games (45) with both tags. 439 tags → **80 nodes, 177 edges**. |

## 5 The picture




**Survival Horror links #1's tags: 2 steps apart, 5 without it.** 14 of 80 tags: ranked pairs (bold), #1 and #2's top-2 neighbours, linking paths. Size = games, width = NPMI. *Not shown. 2025 check = crowding · reviews.

**Selection rule:** of pairs on ≥73 games (0.8%), keep those **not crowded** (games ≤ 1.5× chance; chance = games with A × games with B ÷ 9,094) with ≤20% **shared neighbours**; rank by median reviews ÷ 40.

| Rank: pair | Games (chance) | Crowding | Shared neighbours | Median reviews | 2025 check |
|---|---|---|---|---|---|
| #1: **First-Person × Multiplayer** | 78 (71) | 1.10× | 8% | 204 (5.1×) | 1.14× · 2.4× |
| #2: **Horror × Multiplayer** | 114 (93) | 1.23× | 8% | 117 (2.9×) | 1.45× · 2.7× |
| #3: **Pixel Graphics × RPG** | 133 (107) | 1.25× | 0% | 70 (1.8×) | 1.27× · 1.0× |
| Crowded*: **Building × Simulation** | 74 (29) | 2.54× | 8% | 136 (3.4×) | 2.46× · 2.5× |

## 6 Findings

- **#1** is near chance (NPMI 0.02, no edge) yet gets 5.1× the median reviews ($p < 0.001$, permutation test); 68% are horror.
- **Survival Horror is the only link** between #1's two neighbourhoods (figure): the natural third tag.
- **Holds up:** #1 at all 8 edge cutoffs; above the median in 100/100 bootstraps (#1 in 59); 3.4× without its top 5 games; passes the 2025 check (table).

## 7 Recommended action

- Week 0: pitch a first-person multiplayer horror game.
- Weeks 1–6: prototype. Week 7: Coming Soon page led by First-Person and Multiplayer/Online Co-Op, with Survival Horror third.
- Every quarter: re-run on new releases (section 8).

## 8 What would change my mind

#1 and #2 are one multiplayer-horror bet, and First-Person is spreading (27% of 2025 games). If #1 passes 1.5× crowding, I switch to #3 (only 1.0× in 2025).